

BROOKE WINKLER

MECHANICAL ENGINEERING · AEROSPACE ENGINEERING

B.S. Mechanical Engineering & Aerospace Engineering · Honors Program · Michigan Technological University · Expected April 2030

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Seeking a Summer 2027 motorsports engineering internship

Mechanical and Aerospace Engineering student at Michigan Technological University (Honors Program, Class of 2030) with four years of FIRST Robotics design, fabrication, electrical, and leadership experience, including a FIRST Impact Award and World Championship qualification. Independent research into the 2026 Formula 1 power unit regulations and their effect on racing.

RACE OPERATIONS

01

Captain — Design, Strategy, Build & Electrical Lead

Aug 2022 – May 2026

FRC TEAM 7598 — SCA CONSTELLATIONS · WIXOM, MI

- Ran match strategy, scouting and real-time pit decisions through six-week build-and-compete seasons on a rookie-heavy roster.
- Tuned robot parameters and operational values between and during matches — hands-on telemetry work that made the calls I made executable.
- Owned the technical chain end to end: SolidWorks design, CNC / 3D-print / laser fabrication, wiring, and Java + WPILib bring-up.
- Mentored newer members into independent ownership of design, build, pit and match operations.
- FIRST Impact Award · World Championship qualifier · regional & state competition runs

Media Team — Photographer & Journalist

Jul 2026

RED BULL RACING DETROIT SHOWRUN · DETROIT, MI

- Worked inside a top-tier motorsports event weekend, supporting media and race-adjacent operations and seeing how a professional paddock is run.
- Filed coverage with local news outlets — turning raw event detail into a story a wider audience could follow.

Business Development

Dec 2025 – Present

AVEXEL WEB DESIGN · HARBOR SPRINGS, MI

- Built the outbound engine from nothing — process, materials, qualification criteria and client pipeline.
- Helped generate the revenue that funded an entire FRC competition robot.

STRATEGY & SIMULATION

02

2026 Formula 1 Power Unit Regulations — Independent Research

2026 – Present

POWERTRAIN, DEPLOYMENT & COMPETITIVE DYNAMICS · SELF-DIRECTED

- Reading the 2026 PU regulations clause by clause: the near-50/50 ICE-to-electric split, deployment modes, sustainable fuels, active aero.
- Mapping each rule to its on-track consequence — deployment strategy, race rhythm, overtaking mechanics, and how the field reshapes.
- Writing it as a briefing I would defend line by line — the discipline I want to apply to real team data.

LEADERSHIP UNDER PRESSURE

03

Varsity Golf Captain

2022 – 2025

ST. CATHERINE OF SIENA ACADEMY · WIXOM, MI

- Senior-year captain after three varsity seasons — reading conditions, committing to a plan, holding rhythm under a clock.
- All-Catholic · All-League · All-Academic

Technical Director & Stage Manager

2023 – 2026

SCA DRAMA · WIXOM, MI

- Ran backstage technical operations and crews across productions, solving logistics and timing problems live.
- 2x Superior Award — Stage Management

Mentor & Volunteer

2022 – 2026

FIRST ROBOTICS — SOUTHEAST MICHIGAN · MICHIGAN

- Mentored two FTC teams, including standing up a rookie organisation and coaching one to the Michigan State Championship.

BY THE NUMBERS

4 seasons of competition operations

6 week build season, every year

125 lb competition robots, CAD to floor

2 FTC teams mentored

EDUCATION

Michigan Technological University

B.S. Mechanical Engineering & Aerospace Engineering

Houghton, MI

Honors Program · expected April 2030

St. Catherine of Siena Academy

High School Diploma

Wixom, MI

GPA 3.8 · May 2026

TOOLKIT

DATA & SIMULATION

MATLAB · Mathematica · Parameter tuning · Data analysis · Scouting & pattern reading

RACE OPERATIONS

Match strategy · Pit operations · Event operations · Technical briefing · Team leadership

CAD & DESIGN

SolidWorks · Onshape · Fusion 360 · AutoCAD

FABRICATION

3D printing · CNC machining · Laser cutting · Soldering · Mechanical fabrication

CONTROLS & CODE

Java · WPILib · JavaScript · Git / GitHub

HONORS

- 3x Top Technician in the State of Michigan
- NASA-funded scholarship recipient
- Renaissance Woman Award
- FIRST Impact Award · World Championship Qualifier

COURSEWORK

FIRST-YEAR, MICHIGAN TECH

Calculus I & II · University Physics · General Chemistry · Engineering Fundamentals · Technical Graphics / CAD

NEXT

Formula 1, IndyCar, IMSA or an OEM team — anywhere strategy and telemetry are taken seriously. Building toward Formula SAE at Michigan Tech and eager to put hands on a real data stack.

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